

Multiscale Model for Spatial Cell–Cell Interactions

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Motivation

Spatial organization provides clues about potential cell–cell interactions [1], which we characterize through spatial associations between cell populations.

- Cell–cell associations can **change with spatial distance**.
- Existing approaches often summarize spatial associations using **distance bins or fixed neighborhoods**.
- These summaries can **obscure distance-dependent patterns**, particularly associations that change direction or occur only at specific spatial scales.
- Goal:** characterize how spatial associations between cell populations **vary continuously with distance**.

Existing approaches

	Co-occurrence	Neighborhood enrichment
Example	<i>Squidpy</i> [2]	<i>Giotto</i> [3]
Measure	Enrichment of cell type B around A across distance bins	Enrichment of A–B pairs within a defined neighborhood
Concept	$\frac{\text{Observed proportion of B around A}}{\text{Global proportion of B}}$	$\frac{\text{Observed A–B neighboring pairs}}{\text{Expected A–B pairs under permutation}}$

Limitation Provides distance-dependent summaries, but **does not directly quantify uncertainty** (e.g., confidence intervals) for the association across distance

Provides an aggregate neighborhood summary, but can **obscure interactions that occur at specific spatial scales or change direction with distance**

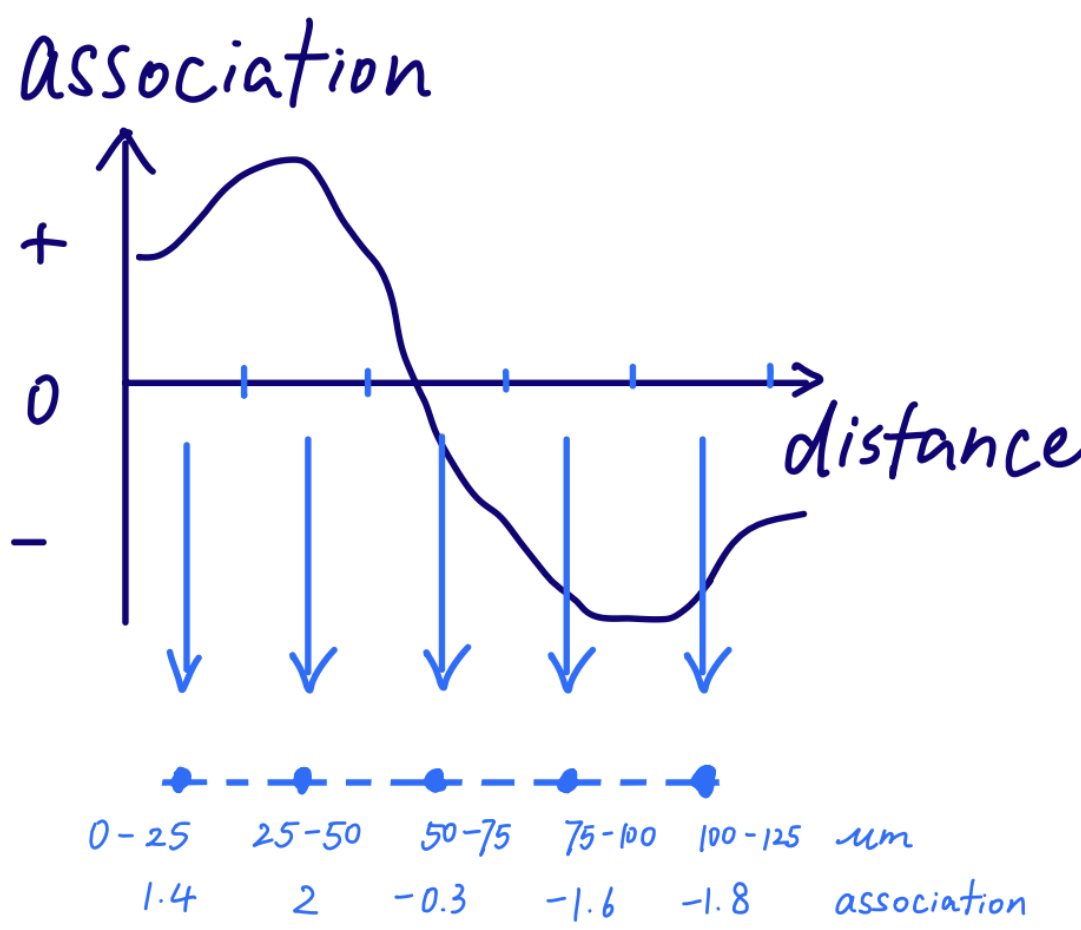


Figure 1. Distance-bin summarization

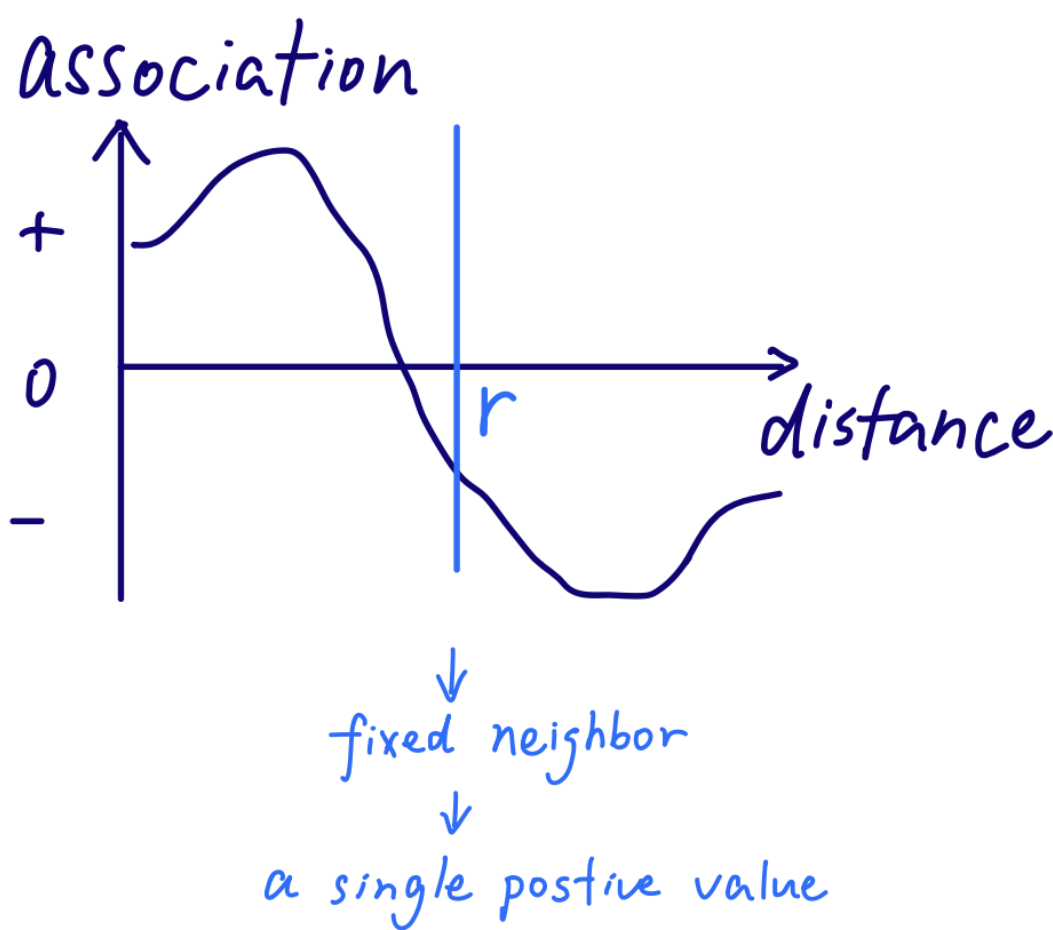


Figure 2. Fixed-neighborhood summarization

Need:

Continuous distance resolution

Flexible interaction patterns

Uncertainty quantification

→ MuCCI

Our method: MuCCI

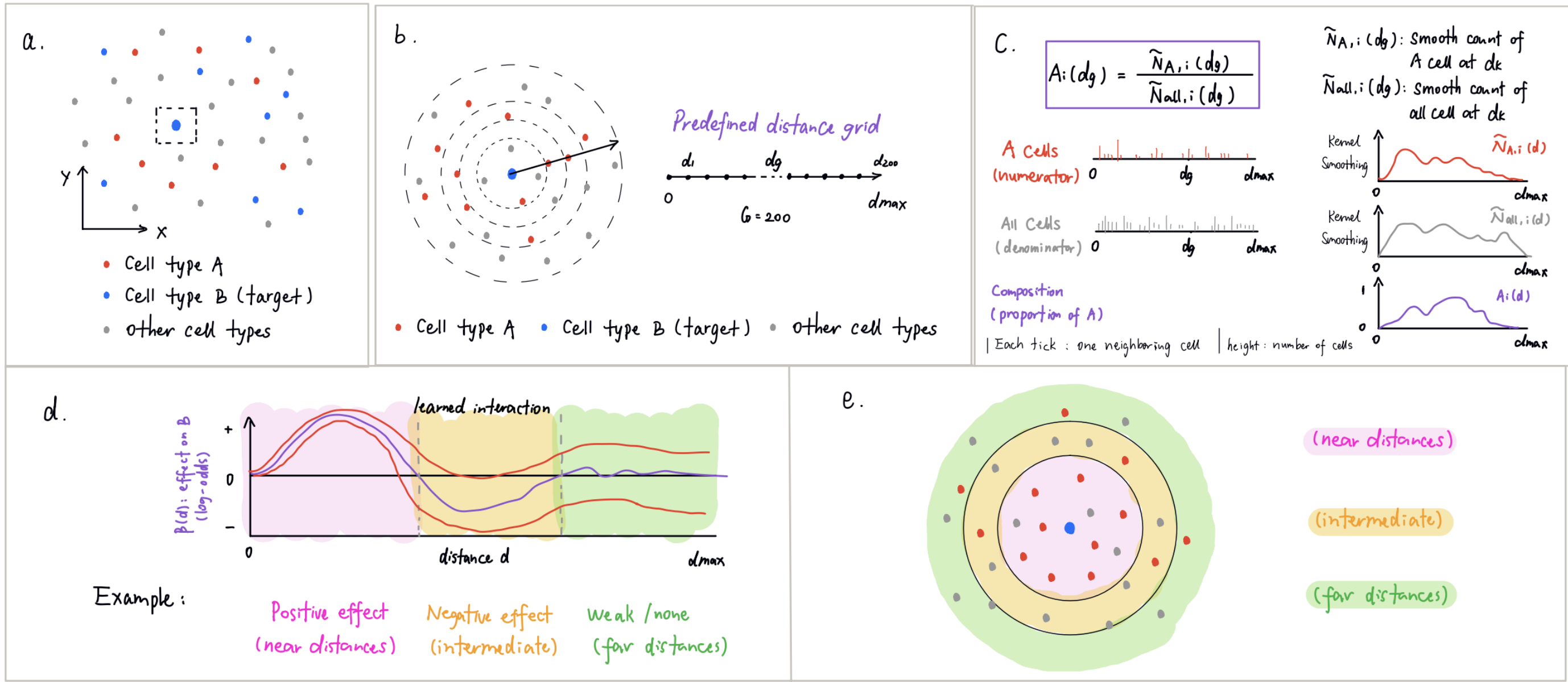


Figure 3. Workflow

- a. **Spatial cell map.** Each cell is characterized by its **cell type** and **spatial location**.
- b. **Distance-resolved neighborhood.** For each center cell i , surrounding cells are evaluated over a predefined distance grid.
- c. **Local composition profile** $A_i(d)$. The local proportion of neighboring type-A cells is represented as a smooth function of distance.

MuCCI relates this neighborhood profile to the identity of the center cell:

$$\logit\{P(Y_i = 1)\} = \beta_0 + \int A_i(d)\beta(d)dd$$

where $Y_i = 1$ if center cell i is type B.

- d. **Output: distance-dependent interaction effect** $\beta(d)$. $\beta(d)$ represents the distance-dependent interaction effect between neighboring type A cells and center-cell type B. Confidence intervals quantify uncertainty across distance.

- $\beta(d) > 0$: Positive interaction
- $\beta(d) = 0$: No interaction
- $\beta(d) < 0$: Negative interaction

- e. **Spatial interpretation.** The estimated profile identifies **positive, negative, or weak/no associations** at near, intermediate, and far distances.

Simulations

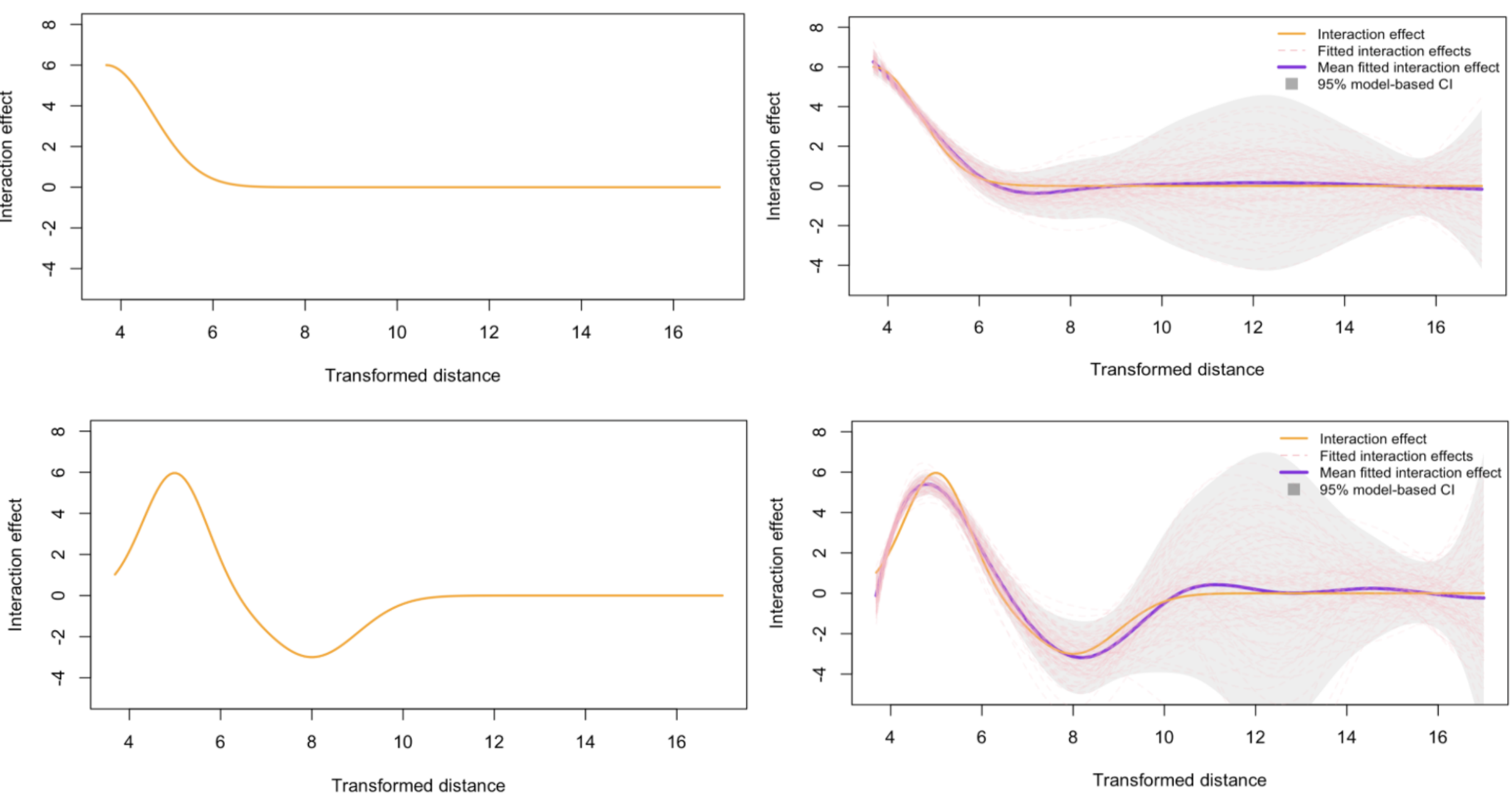


Figure 4. MuCCI recovers monotonic and non-monotonic interaction patterns. **Left:** specified true interaction effects used to generate the simulated data. **Right:** interaction effects estimated by MuCCI across simulations.

Benchmarking

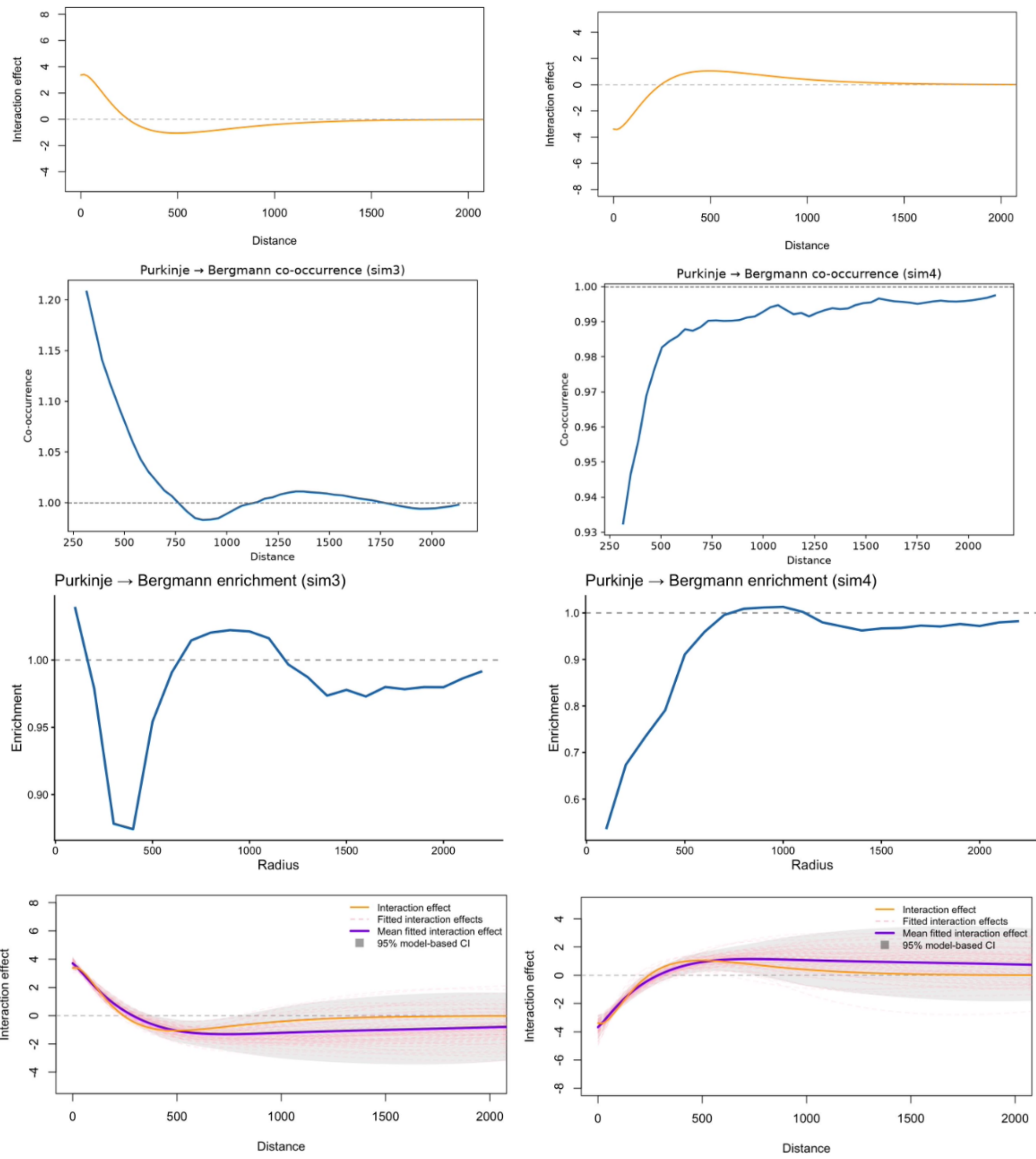
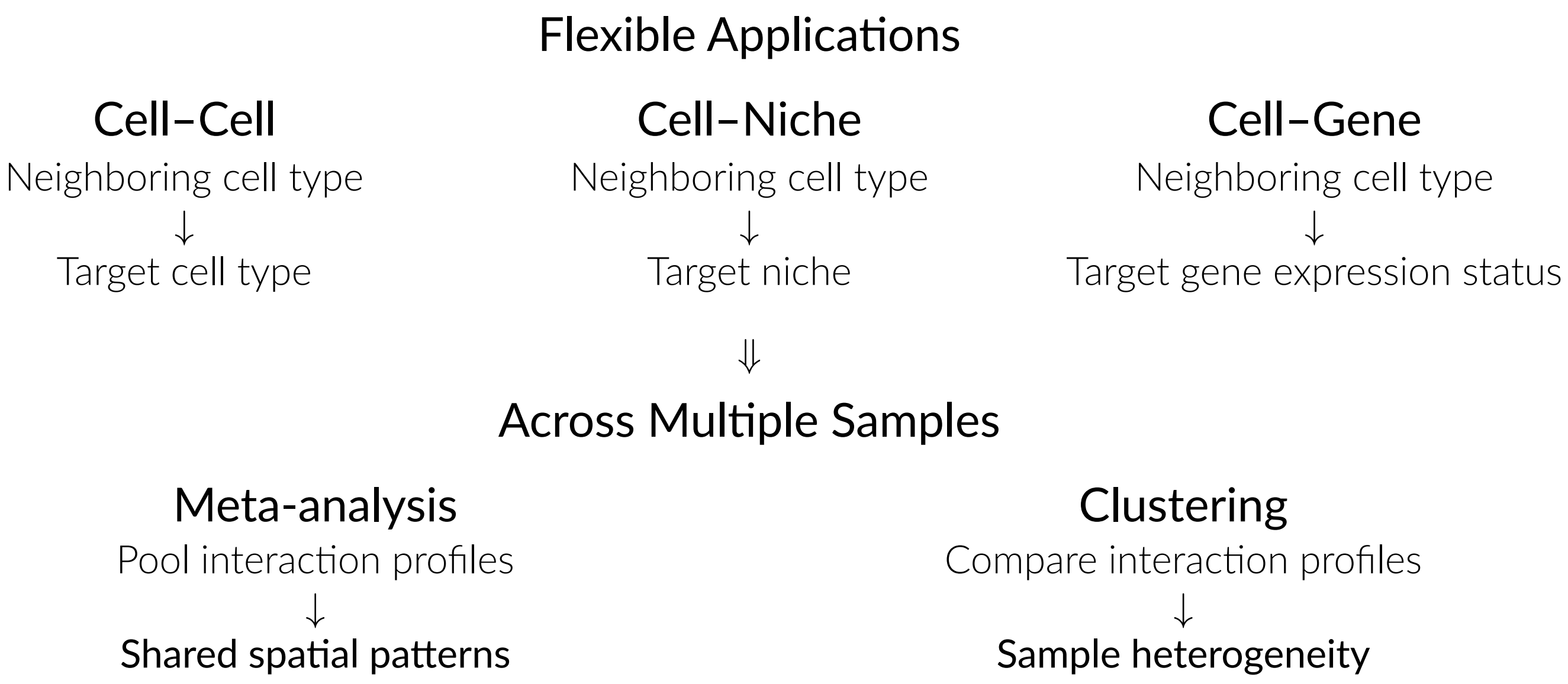


Figure 5. MuCCI compared with Squidpy co-occurrence and Giotto neighborhood enrichment across simulated interaction patterns. **Row 1:** specified true interaction effects; **Row 2:** Squidpy co-occurrence; **Row 3:** Giotto neighborhood enrichment; **Row 4:** MuCCI estimates.

Applications and Downstream Analyses



References

[1] Lu Tian, Fei Chen, and Evan Z. Macosko. The expanding vistas of spatial transcriptomics. *Nature Biotechnology*, 41:773–782, 2023.

[2] Giovanni Palla, Hannah Spitzer, Michal Klein, et al. Squidpy: a scalable framework for spatial omics analysis. *Nature Methods*, 19:171–178, 2022.

[3] Ruben Dries, Qian Zhu, Rui Dong, Chee-Huat Linus Eng, Huipeng Li, Kan Liu, Yuntian Fu, Tianxiao Zhao, Aniruddh Sarkar, Feng Bao, Rani E. George, Noah Pierson, Long Cai, and Guo-Cheng Yuan. Giotto: a toolbox for integrative analysis and visualization of spatial expression data. *Genome Biology*, 22(1):78, 2021.